

e-CAM_YUV-GMSL

Release Package Manifest



Version 1.0

e-con Systems

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Introduction to e-CAM_YUV-GMSL

e-con Systems is a leading Embedded Product Design Services Company which specializes in advanced camera solutions. e-CAM_YUV-GMSL supports four products NileCAM20_CUOAGX, NileCAM21_CUOAGX, NileCAM25_CUOAGX and NileCAM81_CUOAGX.

NileCAM20_CUOAGX is a 2 MP GMSL camera and the successor product of the NileCAM family of Serial Link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0230 CMOS image sensor from onsemi® along with on-board ISP for the camera module board and GMSL SerDes interface board.

NileCAM21_CUOAGX is a 2 MP GMSL camera is the successor product of the NileCAM family of serial link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0233 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

NileCAM25_CUOAGX is a 2.3 MP GMSL camera and is the successor product of the NileCAM family of serial link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0234 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

NileCAM81_CUOAGX is an 8 MP Gigabit Multimedia Serial Link (GMSL) camera and is the successor product of the NileCAM family of Serial Link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0821 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

This document explains the directories of package manifest and the contents of release package for Jetson AGX Orin™ development kit.

Package Manifest

This section describes the detailed information of package manifest.

The following table lists the top-level directories and files that are created when you extract the

e-CAM_YUV-GMSL-

PRODUCTS_JETSON_AGX_ORIN_<L4T_version>_<release_date>_<release_version>.tar.gz release package.

Table 1: Package Manifest Directories and its Descriptions

Directory	Description
Application/	Directory containing application source code and dependencies.
Kernel/	Directory containing kernel patch file for e-CAM_YUV-GMSL products.
misc/	Directory containing miscellaneous files that supports e-CAM_YUV-GMSL products.
Firmware	Directory containing firmware that supports e-CAM_YUV-GMSL products.
e-CAM_YUV-GMSL-PRODUCTS_JETSON_AGX_ORIN_<L4T_version>_<Jetpack_version>_<release_version>.tar.gz	Compressed archive containing required binaries for updating Jetson AGX Orin™.
install_binaries.sh	Install script for installing prebuilt kernel, application binaries and miscellaneous on the Jetson AGX Orin™ development kit.
release_integrity.md5	Encrypted file containing the md5sum of the e-CAM_YUV-GMSL products archive.

Overview of Package Manifest Directory

This section provides detailed information about each directory of the release package.

Application

The following table lists the directories containing the application prebuilt binaries and source code required for e-CAM_YUV-GMSL products camera on the Jetson AGX Orin™ development kit.

Table 2: Application Directories and its Descriptions

Directory	Description
ecam_tk1_guvcview/Source/	Directory containing application related source code for Jetson AGX Orin™.
ecam_tk1_guvcview/Source/ e-CAM_TK1_guvcview-src-1.7.2.tar.gz	Archive file containing ecam_tk1_guvcview application source code for Jetson AGX Orin™.
e-multicam/Source/	Directory containing e-multicam application related source code for Jetson AGX Orin™.
e-multicam/Source/e-multicam.tar.gz	Archive file containing e-multicam application source code for Jetson AGX Orin™.

Kernel

The following table lists the directories containing the kernel prebuilt binaries and source code required for e-CAM_YUV-GMSL products camera on the Jetson AGX Orin™ development kit.

Table 3: Kernel Directories and its Descriptions

Directory	Description
e-CAM-YUV_GMSL_JETSON_ _ORIN_<L4T_version>_kernel.patch	Patch file related to the default kernel, device-tree, camera module and nvidia-oot for Jetson AGX Orin™ that adds support for e-CAM_YUV-GMSL products.
e-CAM-YUV_GMSL_JETSON_ORIN_<L4T_version>_dtb.patch	
e-CAM-YUV_GMSL_JETSON_ORIN_<L4T_version>_module.patch	
e-CAM-YUV_GMSL_JETSON_ORIN_<L4T_version>_oot.patch	

Misc

The following table lists the directories containing the miscellaneous files that are required to support e-CAM_YUV-GMSL product on the Jetson AGX Orin™ development kit.

Table 4: Misc Directories and its Descriptions

Directory	Description
v4l2-compliance	v4l2-compliance binary shared by Nvidia to run v4l2-compliance test

Glossary

CMOS: Complementary Metal Oxide Semiconductor.

DTB: Device Tree Blob.

GMSL: Gigabit Multimedia Serial Link.

ISP: Image Signal Processor.

L4T: Linux for Tegra.

Support

Contact Us

If you need any support on e-CAM_YUV-GMSL product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	12-Aug-2024	Initial draft	Camera Dev Team